

KATHMANDU UNIVERSITY

Discrete Mathematics (MCSC 201)

Assignment – 3

2026

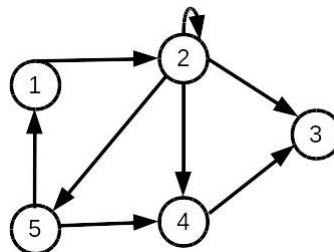
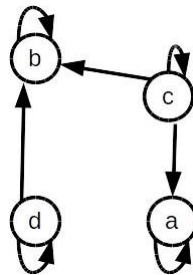
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1. Define a relation and a partition. List all partitions of $S = \{1, 2, 3, 4\}$.
2. Define the equivalent relation. If R and S be the relation From A to B . If $R(a) = S(a)$ for all a in A , Prove that $R = S$.
3. Let $A = \mathbb{R}$. Consider the relation R on A such that $a R b$ if and only if $2a + 3b = 6$. Find the $\text{Dom}(R)$ and $\text{Ran}(R)$.
4. Find the domain, range, matrix, the digraph of the relation R .
 - (a) $A = \{1, 2, 3, 4, 8\}, B = \{1, 4, 6, 9\}; aRb$ if and only if $a|b$.
 - (b) $A = \{1, 2, 3, 4, 8\} = B; aRb$ if and only if $a + b \leq 9$.
 - (c) $A = \{1, 3, 5, 7, 9\}, B = \{2, 4, 6, 8\}; aRb$ if and only if $b < a$.
5. Give the relation R defined on A , its digraph, and give the in-degree and the out-degree of each vertex. Also determine the types of relations.

(a) Let $A = \{1, 2, 3, 4\}$ and $M_R = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

(b) Let $A = \{a, b, c\}$ and $M_R = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

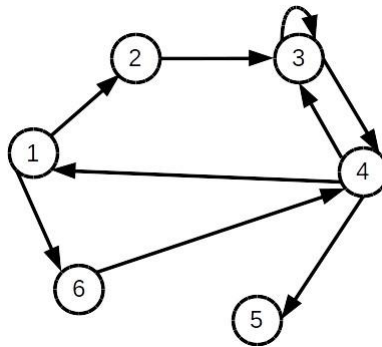
6. Find the relation determined by the digraph and give its matrix. Also determine the in-degrees and out-degrees of each vertex.



7. Let R be the relation whose digraph is given in figure below

- (a) List all paths of length 1.

- (b) List all paths of length 2 starting from vertex 2.
- (c) List all paths of length 3.
- (d) Find a cycle starting at vertex 2.
- (e) Find a cycle starting at vertex 6.
- (f) Draw the digraph of R^2 .
- (g) Find M_{R^2}
- (h) Find R^∞ and M_{R^∞} .
- (i) If $\pi_1 : 1, 2, 3$, and $\pi_2 : 3, 4, 1$, find the composition $\pi_2 \circ \pi_1$. Is the composition a cycle?



- 8. If $S = \{1, 2, 3, 4\}$, $A = S \times S$. Define a relation R on $A : (a, b) R (c, d) \Leftrightarrow ad = bc$. Determine the types of relation.
- 9. Let $S = \{1, 2, 3, 4\}$ and $A = S \times S$. Define the following relation on $A : (a, b) R (c, d) \Leftrightarrow a + b = c + d$. Show that R is an equivalence relation.
- 10. Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3\}$. Given the matrices M_R and M_S of the relations R and S from A to B , compute (a) $M_{R \cup S}$ (b) $M_{R \cap S}$ (c) $M_{R^{-1}}$ and (d) M_S^{-1} .

$$M_R = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } M_S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$